- You use the IBM Spectrum Protect fixed content device for Content Platform Engine storage

Known issues and limitations for content services containers

The following information describes known issues and limitations when you deploy content services containers. These limitations cover the deployment process as well as using the content services components in a container environment.

Support details for content services components in container environments

For details about support in container environments, see the Software Product Compatibility report for the following products:

- FileNet Content Manager V5.5.1, V5.5.2, V5.5.3, V5.5.4
- IBM Content Foundation Container V5.5.1, V5.5.2, V5.5.3, V5.5.4
- IBM Content Navigator Container V3.0.4, V3.0.5, V3.0.6, V3.0.7

Limitations for IBM Content Navigator

(V5.5.3 and earlier) IBM Content Navigator Task Manager

The IBM Content Navigator Task Manager feature and any functions associated with the feature are not supported when IBM Content Navigator is deployed in a container environment. This limitation includes the following functions:

- Teamspace deletion
- IBM Content Navigator Box Share
- IBM Case Manager Box Integration
- IBM Enterprise Records sweep
- IBM Content Collector for SAP

Limitations for Content Platform Engine

EJB transport and any client applications or features that require EJB transported are not supported when Content Platform Engine is deployed in a container environment.

(V5.5.4) Known Issue: zero-byte file causes error message

When you deploy the external share container with the operator, but do not use the external LDAP settings, a zero-byte file called `ibm_ext_ldap_AD.xml` is created inside the container. This file triggers an invalid file error message. To solve the issue, you must manually delete the file from the disk.

(V5.5.1) Known Issue: The Initialization and Verification options in the configuration tool are not working properly for environments that are using Oracle databases

If you are using Oracle database configured through the IBM Business Automation Configuration Container, do not select the optional steps Initialize and Verification in the configuration tool. If a user uses Initialize and Verify with an Oracle database the object stores are not created correctly.

By default the IBM Business Automation Configuration Container Initialize option configures shared database connections for the object store. However, this process works only with certain prerequisites in place. The recommended approach is to not use the Initialize and Verify options in the configuration tool so that you don't encounter issues due to the shared database connections without the proper prerequisites.

Workaround:

If you are deploying with an Oracle database, make sure that the following options in the configuration tool are not selected:

- Include initialization step
- Include verification step

When your deployment completes, you must initialize your Content Platform Engine and other components manually. See the following topic for details: "Initializing and verifying your content services environment" on page 112

If you selected the options during configuration, you can fix the environment by deleting the created object store, cleaning the associated tablespace, and recreating the object stores manually through the Administration Console for Content Platform Engine.

(V5.5.3 and earlier) Known Issue: When using IE or Edge browser to access the IBM Business Automation Configuration Container tool, the log output might not display as expected

On the deployment result page, the job container displays the log output in real time. When using an Internet Explorer or Edge browser, sometimes no log output displays and the deployment progress continues to show as not started.

The EventSource API that is used to fetch the server log information is not supported by Internet Explorer and Edge browsers, as explained in the following information: https://developer.mozilla.org/en-US/docs/Web/API/EventSource#Browser_compatibility

Workaround:

Use other browsers like Chrome, Safari, and Firefox.

Known Issue: In some cases the rerun function displays the log of the previous event before starting with the new event information

When a job fails, after you troubleshoot and fix the issues, you can rerun the failed job by clicking **Rerun** in the user interface.

When you click Rerun, it can take some time for the backend job container to restart and clean the log of the previous run. It can happen that the log is requested before the previous log is cleaned, and in this situation it prints the older log.

Resolution:

To get a clean view, after the new log starts to display messages, refresh the page. The old log is removed from the display.

(V5.5.1) Known Issue: Initializing IBM Content Navigator

When you choose the **Include initialization step** while deploying IBM Content Navigator, the application initializes a repository and a new default desktop.

Known Issue: When you try to edit the repository, the **Save** button is disabled.

Reason: The repository cannot be saved because the default repository ID includes an underscore ("_").

Impact: You cannot modify the repository settings. However, there is no impact to the functionality connecting to the object store in FileNetP8. Users still can browse, create documents, and add documents to the repository.

Resolution: Update the following fields in the `jobs.yml` file to values that do not include an underscore:

- ADD_REPO_ID
- ADD_DESKTOP_REPO_ID

Chapter 2. Planning for container deployment

Depending on your starting point, your actions to prepare for and deploy your container environment can vary. Consider the scenario that best matches your goals for the container deployment.

The preparation steps and deployment process are different depending on both your starting point and your goal for the container environment. Although the component applications are deployed in the container environment, in most cases the environment still requires many of the same FileNet P8 infrastructure components and preparation steps that are needed in an on-premises installation.

Review the roadmaps for the following deployment scenarios to determine the steps you need to take to prepare for and run the container deployment:

Creating an all-in-one developer environment
: This scenario works well for a demonstration server or non-production developer workstation. Because this complete platform relies on open LDAP and a generic database configuration, this installation method is not appropriate for production-level use. You can use the all-in-one installation tool to create environments for demonstration or testing purposes, or to try out the FileNet P8 system in a container environment before you move to containers.

Creating a new instance of FileNet P8 with container deployment
: This scenario assumes that you are creating an entirely new environment for FileNet P8. To prepare for this environment, you must follow the same planning and preparing topics as you would for an on-premises installation, with some exceptions. For example, you must create a database, configure your LDAP system, create users, and take other steps to prepare before you are ready to deploy the component containers.

Adding container instances to an existing FileNet P8 domain
: In this scenario, you already have an on-premises installation of FileNet P8 running. You are either adding container instances of the applications alongside your existing system, or moving from your existing on-premises system to one in which your components are deployed as containers.

  Because you already have a working FileNet P8 environment, you do not need to set up your database, LDAP server, users, and so on. However, you must still take steps to prepare your environment and the container environment for deployment.

Roadmap for a new FileNet P8 domain in a container environment

You prepare your FileNet® P8 infrastructure components as well as the servers and configuration settings that are required for the container environment, then use the method you choose to deploy the component containers.

FileNet® P8 requires a robust environment of integrated software, such as databases, Lightweight Directory Access Protocol (LDAP) servers for access configuration, containers, and multiple required or optional server and client components in the FileNet P8 family. This interdependence calls for preparation and collaboration with the administrators in your environment to configure the infrastructure and record the relevant details for the FileNet P8 deployment and setup.

The following roadmap presents a version of the preparation and deployment steps for your FileNet P8 setup process:

1. Read the whole Planning and Preparing for FileNet P8 section, and make the following decisions:

   - Which administrators and team members in your organization to designate for the different administrative roles and tasks
   - Which installation scenario is most relevant and useful for your needs

   Note: Because a web application server is included in the component container deployment, preparation steps for application server settings do not apply.

2. Complete the tasks to prepare for deployment. These tasks are likely to be performed by several different administrators. Create a collaborative record to gather all the settings you need for the IBM

Business Automation Content Services Configuration tool or YAML file that you run to perform the deployment.

- Configure storage for the FileNet P8 domain.
- Set up the directory server accounts for all of the roles and users your environment requires.
- Create databases to contain the Content Platform Engine GCD and object stores. You can create and deploy up to 2 object stores as part of the container deployment process.
- If you plan to include the IBM Content Navigator container, do the following additional steps:
  - Set up the directory server accounts for all of the roles and users your IBM Content Navigator environment requires.
  - Create a database to contain the IBM Content Navigator configuration information.
- Prepare the server instance that will host the content services applications. For example, for IBM Cloud Private you might create a distinct user and namespace to facilitate the management of the content services deployed components.
- Set up the worker nodes where the component containers will be deployed. This includes preparation of shared storage for configuration information, logs, and the document content and optional full text indexes, along with associated persistent volumes and persistent volume claims.

3. Make the component container images available on the IBM Cloud Private or Kubernetes server.
4. Use the settings recorded by your setup administrators to complete the configuration tool fields or YAML file, and run the deployment. This step deploys your applications and prepares them for use.

Roadmap for a container deployment with an existing FileNet P8 domain

When you deploy containers to work with an existing FileNet P8 domain, your environment infrastructure is already set up. You prepare the environment by upgrading to the appropriate version.

Upgrade planning depends on the details of your existing environment. The starting version and platform choices all influence the upgrade path of your existing components.

It is recommended to first upgrade the WebSphere Application Server on-premises system to the targeted level of FileNet Content Manager. Run the FileNetCPE Engine with its associated client applications to ensure the system is operating as expected. Then move the FileNet P8 domain to be managed by a container based deployment.

However it is supported to both upgrade and move at the same time.

To upgrade to Content Platform Engine V5.5.x, your system must be at V5.2.0 or later. If you want to move your environment to a deployment on a Certified Kubernetes platform, your FileNet P8 domain must be at V5.5.1 or later. Use the following information to complete that initial upgrade step: Planning and preparing for FileNet P8 upgrade.

The following roadmap presents a version of the preparation and deployment steps for your FileNet P8 setup process:

1. Read the Upgrade scenarios section, and make the following decisions:
   - Will the current WebSphere® Application Server hosted deployment of the FileNetEngine application continue to be used in a side-by-side configuration with the Content Platform Engine container deployment?
     - If yes, then the WebSphere Application Server hosted deployment must be upgraded to the same version of FileNet Content Manager as the container-hosted deployment. Plan to upgrade the WebSphere Application Server hosted environment first, to Content Platform Engine V5.5.2 or later. After the upgrade, use the following information to add containers to an existing P8 Domain that is managed by WebSphere Application Server: “Adding containers to an existing on-premises P8 domain (side-by-side deployment)” on page 119
     - If no, then plan to shut down the existing WebSphere Application Server hosted environment and use the following information to upgrade an existing P8 Domain using containers only: “Upgrading an existing P8 domain to a containers-only environment” on page 120

Note: All data in the existing global configuration database, object stores, and workflow system are automatically upgraded when you deploy the Content Platform Engine container.

- Will it be necessary to first upgrade or migrate any underlying vendor software supported by Content Platform Engine? See the following topic for more information: Upgrading or migrating the underlying vendor software supported by Content Platform Engine.

Note: Because a web application server is included in the component container deployment, preparation steps for application server settings do not apply.

2. Complete the tasks to prepare for deployment. These tasks are likely to be performed by several different administrators. Create a collaborative record to gather all the settings you need for the IBM Business Automation Content Services Configuration tool or YAML file that you run to perform the deployment.
   - Ensure that the directory server is supported by the new Content Platform Engine version.
   - Prepare the database server that contains the Content Platform Engine GCD and object stores for upgrade.
   - If you plan to include the IBM Content Navigator container, prepare the database that contains the IBM Content Navigator configuration information for upgrade.
   - Prepare the server instance that will host the content services applications. For example, for IBM Cloud Private you might create a distinct user and namespace to facilitate the management of the content services deployed components.
   - Set up the worker nodes where the component containers will be deployed. This includes preparation of shared storage for configuration information, logs, and the document content and optional full text indexes, along with associated persistent volumes and persistent volume claims.
3. Make the component container images available on the IBM Cloud Private or Kubernetes server.
4. Use the settings recorded by your setup administrators to complete the configuration tool fields or YAML file, and run the deployment. This step deploys your applications and prepares them for use.

(Optional) Upgrade with migration to containers using a staging environment

An upgrade can be accomplished while also migrating from an on-premises environment to a container environment. Making such a change is often part of the motivation for doing the upgrade and it is important to have a well-understood process.

This topic applies only if you want to use a replica or staging environment during the upgrade to minimize the impact of the upgrade on the production system.

Upgrading large FileNet P8 systems involves significant work. The upgrade can be particularly challenging if you are changing the underlying platform of major system components, such as Content Platform Engine. Using this approach, you might install and configure a new server instance, such as for the database server. The initial installation and configuration work can be done without impacting the production system.

At a high level, complete the upgrade migration procedures by using the following steps. Some steps are repeated for each major FileNet P8 component:

- Determine a time when you can run the upgrade, which must be done when nobody is altering the production system data. The copy of the production data (replica) must reflect the production system. Otherwise the upgrade is not on current data.
- Set up a second system that contains a copy of production data. With this approach, you can revert to the original system if you encounter problems during the upgrade. You can also do some of the initial installation and configuration without impact to the production system. This second system lets you move to different server instances, replacing or updating hardware for dependent systems such as directory servers or database servers. Try to reuse as many of the configuration settings as possible from the original system on the second system to reduce any configuration issues that might arise in the upgrade.
- On the second system, run all upgrade tasks that might alter data in a production system.